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ORIGINAL RESEARCH

Deep learning based diagnosis of Parkinson's Disease using diffusion magnetic resonance imaging

Hengling Zhao[1] · Chih-Chien Tsai[2,3] · Mingyi Zhou[1] · Yipeng Liu[1] · Yao-Liang Chen[4,5] · Fan Huang[6] · Yu-Chun Lin[3,4] · Jiun-Jie Wang[2,3,5,7]

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Abstract

The diagnostic performance of a combined architecture on Parkinson's disease using diffusion tensor imaging was evaluated. A convolutional neural network was trained from multiple parcellated brain regions. A greedy algorithm was proposed to combine the models from individual regions into a complex one. Total 305 Parkinson's disease patients (aged 59.9±9.7 years old) and 227 healthy control subjects (aged 61.0±7.4 years old) were enrolled from 3 retrospective studies. The participants were divided into training with ten-fold cross-validation (N = 432) and an independent blind dataset (N = 100). Diffusion-weighted images were acquired from a 3T scanner. Fractional anisotropy and mean diffusivity were calculated and was subsequently parcellated into 90 cerebral regions of interest based on the Automatic Anatomic Labeling template. A convolutional neural network was implemented which contained three convolutional blocks and a fully connected layer. Each convolutional block consisted of a convolutional layer, activation layer, and pooling layer. This model was trained for each individual region. A greedy algorithm was implemented to combine multiple regions as the final prediction. The greedy algorithm predicted the area under curve of 94.1±3.2% from the combination of fractional anisotropy from 22 regions. The model performance analysis showed that the combination of 9 regions is equivalent. The best area under curve was 74.7±5.4% from the right postcentral gyrus. The current study proposed an architecture of convolutional neural network and a greedy algorithm to combine from multiple regions. With diffusion tensor imaging, the algorithm showed the potential to distinguish patients with Parkinson's disease from normal control with satisfactory performance.

Keywords Deep Learning · Convolution neural network · Diffusion tensor imaging · Differential diagnosis · Idiopathic Parkinson's disease

Hengling Zhao and Chih-Chien Tsai contributed equally to this work.

✉ Yipeng Liu
yipengliu@uestc.edu.cn

✉ Jiun-Jie Wang
jwang@mail.cgu.edu.tw

1 School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China

2 Healthy Aging Research Center, Chang Gung University, Taoyuan, Taiwan

3 Department of Medical Imaging and Radiological Sciences, Chang Gung University, Taoyuan, Taiwan

4 Department of Medical Imaging and Intervention, Chang Gung Memorial Hospital at Linkou, Taoyuan, Taiwan

5 Department of Diagnostic Radiology, Chang Gung Memorial Hospital at Keelung, Keelung, Taiwan

6 Department of Medical Physics and Biomedical Engineering, University College London, London, UK

7 Institute for Radiological Research, Chang Gung University, Taoyuan, Taiwan

Introduction

Parkinson Disease (PD) is a primary progressive neurodegenerative movement disorder, characterized with bradykinesia and tremor (Calabresi et al., 2013; Lu et al., 2016). It is estimated to affect approximately 5% of the population over 85. The diagnosis is usually made by the neurologists who based their decision on judgment from the clinical expressions, following generally accepted criteria. The accuracy can be as low as 53% at the early stage of the disease when the duration is less than 5 years (Meara et al., 1999). It is highly desirable for an early and accurate diagnostic tool such that the accurate intervention would be possible and available to the patients.

Neuroimaging examination such as Magnetic Resonance Imaging (MRI) are often performed but only to rule out concomitant neurological disorders when the potential patients first time search for clinical consultancy. Advanced MRI studies based on diffusion-tensor reported improved diagnostic performance (Lu et al., 2016; Salvatore et al., 2014b). Diffusion tensor imaging has been widely used to explore the white matter abnormality in neurodegenerative diseases (Assaf & Pasternak 2008; Dong et al., 2004). Tensor derived indices, such as mean diffusivity (MD) and fractional anisotropy (FA), might reflect the microenvironmental changes within the surrounding tissue (Basser & Pierpaoli 1996). MD is the rotationally invariant of water diffusion (Clark et al., 2011). FA described the degree of anisotropy of the diffusion process (Ahn & Lee 2011). Because the analysis of tensor derived indices usually based on the manual delineation of pathogenesis related regions such as basal ganglia. This procedure can be tedious and subjective and therefore not practical in clinical routine. An automatic procedure for diagnosis could be highly desirable in a clinical environment.

Using artificial intelligence to improve the image based diagnosis has been proposed in tumour (Pereira et al., 2016) and Alzheimer Disease (Islam, 2019). The diagnosis of PD can be a classification task. A series of machine learning methods have been developed for this purpose (Castellazzi et al., 2020; Zacharaki et al., 2009). Support vector machines combined with principal components analysis showed an accuracy of 90% for the diagnosis of patients with clinically diagnosed PD and progressive supranuclear palsy (Salvatore et al., 2014a). The recent advances in deep learning have achieved outstanding performance in image classification tasks (He et al., 2016; Krizhevsky et al., 2012; Schmidhuber, 2015; Sun et al., 2017).

Because only a binary result (e.g., patient or healthy) was output as the final decision, it is often difficult to determine which part of the brain that was involved in the diagnosis. Furthermore, if trained with a limited number of labeled data (Bartlett et al., 1998; Bengio et al., 2017; Caruana et al., 2001; LeCun et al., 2015), the deep learning algorithm might suffer from overfitting problems, where they might contain more parameters than what can be justified by the data. We therefore proposed an algorithm for computer aided diagnosis, in which the brain was parcellated into multiple regions and used for the subsequent training.

The aim of our study is to test the possibility of the proposed algorithm as applied in a clinical scenario. Diffusion tensor imaging was acquired retrospectively from 3 studies. A convolutional neural network (CNN) was implemented and applied to the parcellated brain regions. A greedy algorithm was developed to identify the combinations of the regions of interest. A final diagnostic decision was given by selective stacking. The result was further validated on the blind datasets.

Methods

Images were collected from previous PD studies of prospective design. The study is complied with the declaration of Helsinki. The retrospective use of these data has been approved by the same institutional review board.

Participants

In total, 305 right handed PD patients (M/F = 173/132, averaged age = 59.9±9.7 year old) were included between Sep. 2008 - Dec. 2017 from the neurology clinics of Chang Gung Memorial Hospital, Linkou. Another 227 healthy control (M/F = 104/123, averaged age = 61.0±7.4 year old) were recruited from the local community. The inclusion criteria for patients are (1) diagnosis of PD according to the criteria by National Institute of Neurological Disorders and Stroke in USA (NINDS), except for the age of onset; (2) acquisition of diffusion tensor imaging on a 3T scanner. The exclusion criteria are moderate to severe dementia, severe dyskinesia, significant major systemic diseases, documented abnormality in the brain by MRI or ^{18}FDG PET studies, history of intracranial operation, significant physical disorder or neuropsychiatric disorder and general MRI exclusion criteria. Table 1 showed the demographic data of the participants.

The clinical severity was evaluated by using Unified Parkinson Disease Rating Scale (UPDRS) (Martínez-Martín et al., 1994). The disease staging was assessed by Modified Hoehn and Yahr Staging (MHY) (Goetz et al., 2004). Both scores are not available from 65 patients. The average UPDRS score and the motor subscale were 36.4±23.6 and 21.3±14.4, respectively. The MHY was as following: stage 1/stage 1.5/stage 2/stage 2.5/stage 3/stage 4/stage 5 = 75/22/34/39/59/25/8.

MRI data acquisition

All images were acquired on a 3T MR scanner (MAGNETOM Trio a TIM system, Siemens, Erlangen, Germany). Contiguous 160 axial T1-weighted images were acquired with magnetization-prepared rapid acquisition gradient echo (MPRAGE) sequence and were used in the subsequent image parcellation. The imaging parameters are as follows: TE=2.63 ms, TR=2000 ms, flip angle= 9°, voxel size=1 × 1 × 1, FOV=224 × 256, matrix size=224 × 256, bandwidth=200. The diffusion weighted images were acquired from 3 different imaging protocols as in Table 1.

Brain parcellation

All data reconstruction was performed followed Tsai et al. (Tsai et al., 2020). The pre-processing procedures for diffusion weighted images datasets included 1). diffusion tensor estimation, 2). co-registration from T1 weighted images to B0 image, 3). normalization of the co-registered T1 images to the standard structural template, 4). inverse transformation of the standard template, and 5). Parcellation for maps of tensor derived index.

Diffusion tensor was reconstructed by using Diffusion Kurtosis Estimator (Tabesh et al., 2011) from diffusion-weighted images. Tensor derived indices used in the study included MD and FA.

MD was calculated by Eq. (1):

$$MD = \frac{(\lambda 1 + \lambda 2 + \lambda 3)}{3} \tag{1}$$

Where the λ1, λ2, and λ3 are the three eigenvalues of the diffusion tensor.

FA was calculated by Eq. (2):

$$FA = \frac{\sqrt{(\lambda 1 - \lambda 2)^2 + (\lambda 1 - \lambda 3)^2 + (\lambda 2 - \lambda 3)^2}}{\sqrt{2(\lambda 1^2 + \lambda 2^2 + \lambda 3^2)}} \tag{2}$$

When $\lambda 1 >> \lambda 2 = \lambda 3$, FA ≈1; when $\lambda 1 = \lambda 2 = \lambda 3$, FA = 0.

To extract the region of interest (ROI) from the tensor derived indices, T1 weighted images were first co-registered to B0 images. The co-registered T1 weighted images were subsequently normalized to a structural template (Montreal Neurological Institute152, (Mazziotta et al., 1995)) using affine transformation. The transformation matrix was subsequently inversed transformed and applied to the Automatic Anatomic Labelling (Tzourio-Mazoyer

Table 1 The parameters of the participants and the associated imaging protocols

	Protocol A	Protocol B	Protocol C	Final
Number of Subjects				
HC	104	-	123	227
PD	169	27	109	305
Sex (Men/Women)				
HC	51/53	-	52/71	103/124
PD	90/79	17/10	66/43	173/132
Age (years)				
HC	58.5±7.3	-	62.1±7.1	59.9±9.7
PD	62.1±7.8	63.7±3.5	56.8±6.1	61.0±7.4
*UPDRS	35.0±22.5	-	39.4±25.9	36.4±23.6
*UPDRS III	20.6±13.6	-	22.8±16.1	21.3±14.4
b value (mm/s^2)	1000/2000/3000	1000	1000/2000	
Repetition time (ms)	5700	8200	7800	-
Echo time (ms)	108	96	83	-
Voxel size (mm^3)	2×2×3	2×2×2	2×2×2	-
Gradient	30	64	64	-
Matrix size	96×96	128×128	128×128	-
FOV (mm)	192×192	256×256	256×256	-
Bandwidth (Hz)	1730	1628	1630	-
Slice number	40	64	64	-
Parallel imaging factor	2	2	2	

The table summarized the parameters for the participants and the details of the associated imaging protocols. HC: healthy control; PD: Parkinson's Disease; UPDRS, Unified Parkinson's disease rating scale; UPDRS III, the motor subscale of UPDRS; FOV, field of view, *UPDRS and UPDRS III are only available from patients

et al., 2002) template. The warped template was used as a mask to obtain ROI from the tensor. Each parcellated ROI was extracted from the individual maps of tensor derived index. The maps of the tensor derived index were parcellated into 116 brain regions accordingly. Because of insufficient image coverage, only 90 cortical and subcortical regions were included in the analysis.

Statistical analysis

Statistical analysis was performed using the SPSS statistical package, version 23.0 (IBM, Armonk, NY, USA). Disease severity was expressed with the disease duration, total and motor (III) subscales of UPDRS. The difference between sex was evaluated with the chi-square test. The differences in age, disease duration, and clinical severity (UPDRS and UPDRS III, if available) among imaging protocols were evaluated by using parametric analysis of variance (ANOVA) followed by *post-hoc* (Least Significant Difference). Two-tailed P values < 0.05 were considered statistically significant. The original participants were randomly divided into two groups: dataset of training and cross-validation (N = 432) and dataset of blind-test (hold-out test, N = 100). The 432 samples were evenly divided into ten parts, nine of which was used for training and one for validation. Thus, the cross validation procedure was then repeated for each fold.

Receiver operating characteristic (ROC) analysis was used to assess the diagnostic performance of each CNN model and for each diffusion tensor derived index. The area under the curve (AUC) of the ROC analysis with the optimal cut off points was used to assess the overall discriminative ability of a variable for each ROI.

The diagnostic performance between two combinations was examined by using model performance analysis. The diagnostic performance from combinations from fewer regions was compared to one with the highest AUC. The error rate from two combinations was calculated and the difference of its confidence level was examined (Tan et al., 2016). When the z-score was larger than 1.96, which corresponds to a 95% confidence level at a two-tailed normal curve (p: 0.05/2=0.025), the difference between the two combinations was regarded as statistically significant.

For validation, the parameters of CNN from the regions involved in the combination of reduced number was used to calculate the output *Yn* for each subject in the blind dataset. Optimal cutoff in these regions from the training process was used to calculate *Tn* for each subject. The diagnosis performance was evaluated and reported, including true positive rate, true negative rate, positive predictive value, negative predictive value, accuracy and F1 (harmonic mean of precision and sensitivity).

The procedure of the diagnosis by convolutional neural network

The 3D CNNs were adaptively adjusted to the ROIs using a ten-fold cross-validation dataset in the procedure of model training. The optimized combinations of the ROIs for diagnosis were identified by selective stacking. Figure 1 is a visualization of the proposed algorithm.

Training of CNN

For the CNN model h_θ in each ROI, the process of forwarding propagation can be formulated as Eq. 3:

$$Y_n = h_\theta(\mathbf{X}_n) \tag{3}$$

Where θ represents the whole parameters of CNN as obtained from the input and the output of training dataset. $X_n \in R^{R_1 \times R_2 \times R_3}$ is the input of the ROI_n, where R_1, R_2, and R_3 are the height, width and depth of the input data. *Yn* was the output of CNN, which was a value between 0 and 1. The higher the value, the higher the probability of a patient. The AUC was calculated from the average of the ten-fold cross validation.

Each CNN might contain 2-4 basic convolutional blocks, the details of which can be obtained from Fig. 1. Each convolutional block includes 1-2 convolutional layers, the parameters of which is the value of the 3D convolutional kernels. The 3D convolutional kernels (height× width × depth, output channel number) are set in the architecture of CNN model. The kernel size of all the convolutional layers is $2 \times 2 \times 2$. The number of the convolutional kernels is between 64 and 512.

Selective stacking

After the training of CNN, selective stacking was used to estimate the diagnostic performance as combined from multiple regions. For each region, T_n was the binary thresholded result as Eq. 4:

$$T_n = \begin{cases} 0 \text{ if } Y_n < Opt.\text{cutoff}, \\ 1 \text{ otherwise}, \end{cases} \tag{4}$$

where Opt.cutoff is determined from the optimal cutoff with the best AUC in each ROI. The sample was labelled as patients when T_n is 1, and healthy control if 0.

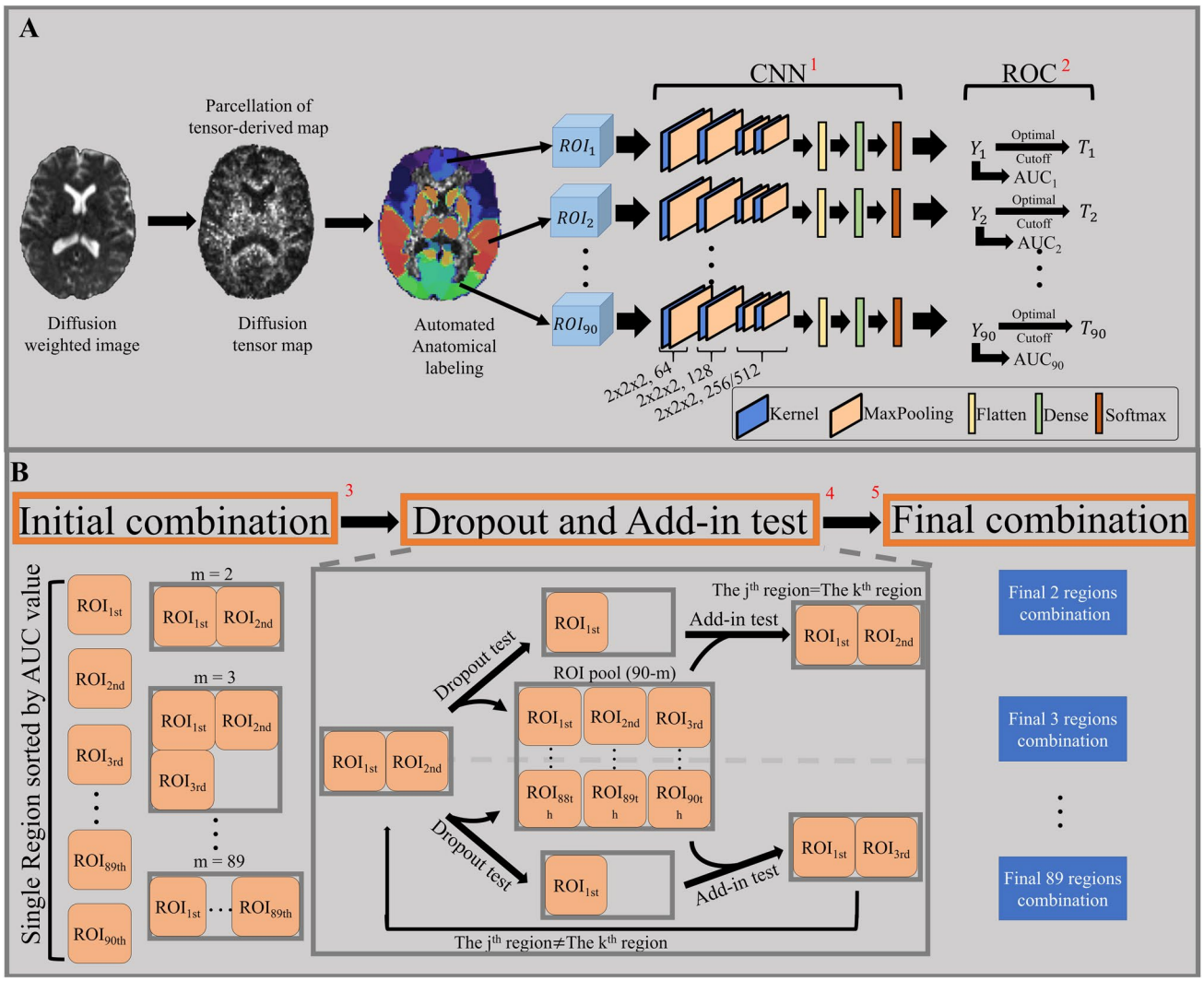

Fig. 1 Diagram of convolutional neural network (CNN) based diagnosis. Figure 1 plotted the preprocessing, the structure of the CNN and the procedure of the greedy algorithm. The number indicated that (1). The architecture of CNN. The kernel size was 2×2×2. The convolution kernels ranged from 64 to 512; (2). The area under curve (AUC) for each region as calculated by receiver operating curve (ROC) analysis, with *Yn* from the corresponded CNN. (3). The first m regions in the initial combination, sorted by AUC in descending order. (4) The processing of dropout and add-in test; The j represented the region contained in the first m regions, $j \in \{1,2,\ldots,m\}$. The k represented the regions not contained in the selected combination, which $k \in \{1,2,\ldots,N\text{-}m\}$, (5). The final combination with the highest AUC

A score S_n of the n-th ROI (ROIn) was calculated by using AUC value from each ROI in the combination of m selected regions as in Eq. 5:

$$S_n = T_n \times \frac{\mathrm{AUC}_n}{\sum_{k=1}^{N} \mathrm{AUC}_k \times A_{m,k}}. \tag{5}$$

The AUC for the n-th region (AUCn), as averaged from ten-fold cross-validation, was calculated by using ROC analysis with *Yn* from CNN of the n-th region. The denominator is the sum of the AUC of ROIs in the selected combination $\mathbb{H}_m$, where m is the number of regions with the selected combination. The score S_n represents the contribution to the diagnosis by the n-th ROI in $\mathbb{H}_m$. $k \in \{1, 2, \cdots, N{=}90\}$ represents a ROI. If $h_\theta^k \in \mathbb{H}_m$, $A_{m,k} = 1$, otherwise 0. $A_{m,k}$ is a binary mask to select the AUC of ROI_k in $\mathbb{H}_m$.

The combination score U_m of selected combination was calculated as in Eq. 6:

$$U_m = \sum_{n=1}^{N} S_n \times A_{m,n}, \tag{6}$$

U_m is used to calculated the AUC of the selected combination. A selective stacking algorithm was implemented to identify the final combinations with m regions ($m = 1, \ldots, N-1$), the process was briefly described as follows:

(1) Initial combination:
The first m regions were selected as an initial combination, which were determined by either AUC sorted in descending order or random selection.

(2) Dropout test:
All selected regions were removed one by one. The AUC from the combination $\mathbb{H}_{m-1}$ was calculated. If removing the jth region resulted in the highest AUC in the selected combination $\mathbb{H}_{m-1}$, where $j \in \{1,2,\ldots,m\}$ represented the region contained in the first m regions, the jth region is discarded.

(3) Add-in test:
All discarded regions were added back one by one. The AUC from the combination $\mathbb{H}_m$ was calculated. If adding the kth region into $\mathbb{H}_{m-1}$ resulted the highest AUC in $\mathbb{H}_m$, where $k \in \{1,2,\ldots,N\text{-}m\}$ represented the regions not contained in the selected combination $\mathbb{H}_{m-1}$, the kth region is added back;

(4) Final combination:
The dropout and add-in tests were repeated for all available combinations. The final combination is when the discarded region is the same as the added one, i.e. (the jth region equal to the kth region).

Results

Figure 2 plotted the AUC values with ten-fold cross-validation from the combination of multiple regions (up to 180 when using both indices). The initial regions were either those with the highest AUC or from random selection. Panel A showed that the highest AUC value in the final combination was 94.1 ± 3.2% in FA (from 22 regions) and 90.4 ± 3.0% in MD (from 41 regions). When using both indices (panel B), the highest AUC value of the combination was

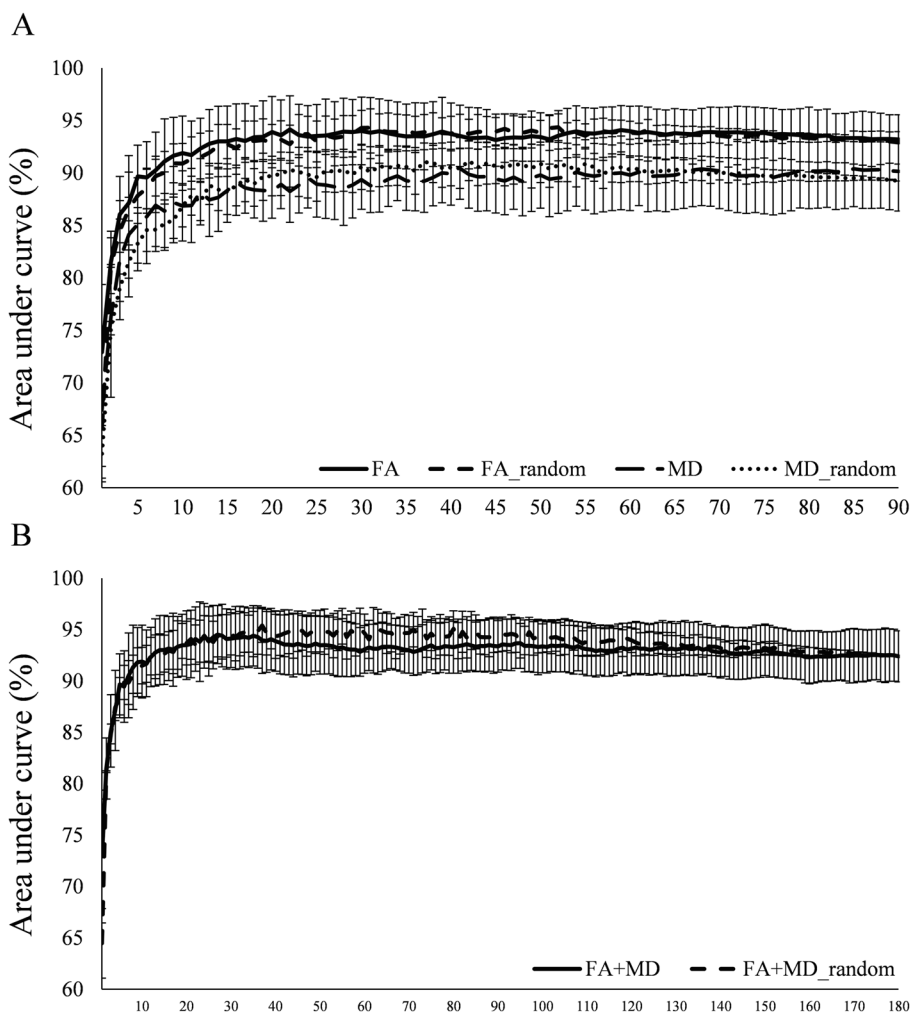

Fig. 2 The averaged area under curve (AUC) with ten-fold cross-validation from the combination of multiple regions by the initial regions with the highest AUC or from random selection. The figure plotted the averaged AUC (in percentage) in the receiver operative curve analysis of fractional anisotropy (FA) and mean diffusivity (MD) in the combination by the greedy algorithm (panel A). Panel B plotted the averaged AUC by using both indices. solid: initial regions from the highest sorted AUC; dashed: initial regions from random selection. The standard deviation was calculated from the ten-fold cross-validation

94.5 ± 3.1% (from 27 regions). No significant difference in diagnostic performance was found between the regions either selected from the highest AUC or randomly.

Figure 3 plotted the AUC in the involved regions composed of the final combination (Panel A: FA 22 regions; and Panel B: MD 41 regions) with ten-fold cross-validation. To compare with the combination with the highest AUC of 22 regions, the difference of the confidence level from all the combinations with fewer number of regions were calculated. We found that the performances were comparable to the highest AUC for combinations from 9 to 21 regions. The same procedure was repeated for MD and in both indices. No statistical significance was noticed in combinations of 12 to 40 regions and 12 to 26 regions when compared to the respective combinations of highest AUC. In summary, the combination of 9 (FA), 12 (MD), and 12 (both) had no statistical significance when compared to the corresponding combinations of highest AUC, respectively.

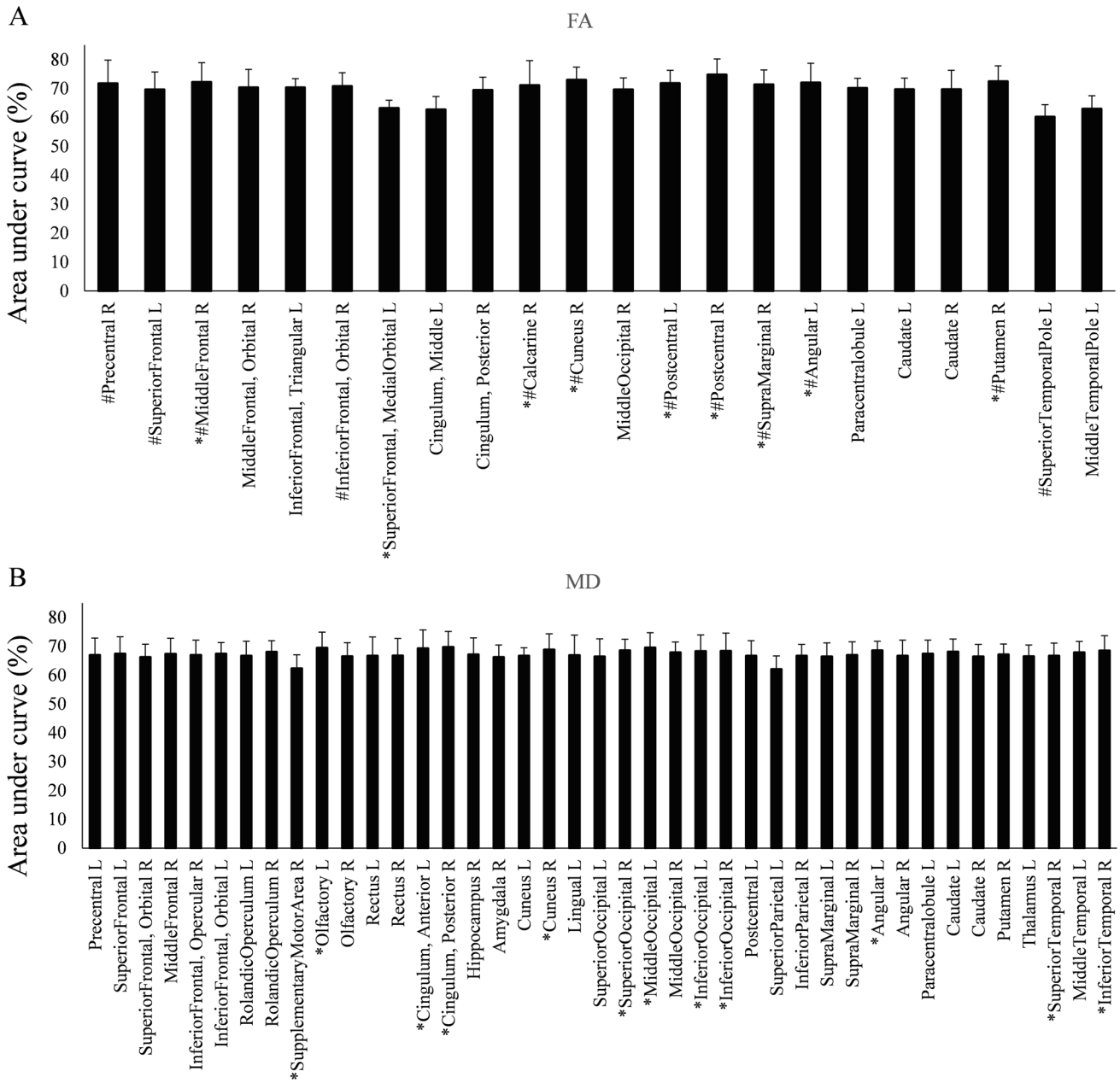

Fig. 3 The averaged area under curve (AUC) in the involved regions composed of the final combination in fractional anisotropy (FA) and mean diffusivity (MD). The figure plotted the AUC (in percentage) in the receiver operative curve analysis of the involved regions in final combination using (**A**) FA and (**B**) MD. The averaged AUC with standard deviation was calculated from the ten-fold cross-validation. * indicated the regions that involved in the combinations with reduced number of regions in FA and MD. # indicated the regions that involved in the combinations with reduced number of regions when using both indices

We further examined the AUC in these involved regions. The overlapped regions in these combinations included right cuneus (AUC: FA, 72.9±4.4%; MD, 68.8±5.5%) and left angular gyrus (AUC: FA, 72.0±6.7%; MD, 68.5±3.2%). The AUC in other involved regions in FA ranged from 63.2±2.6% to 74.7±5.4% (Fig. 3A). The regions included postcentral, putamen, middle frontal gyrus, supramarginal and calcarine in the right hemisphere, as well as medial orbital part of superior frontal gyrus in the left hemisphere.

In MD, AUC in other involved regions was between 62.3±4.7%-69.7±5.4%, which included the posterior cingulate, superior occipital gyrus, inferior temporal gyrus, inferior occipital gyrus, superior temporal gyrus and supplementary motor area in the right hemisphere, as well as middle occipital gyrus, olfactory gyrus, anterior cingulate, and inferior occipital gyrus in the left hemisphere (Fig. 3B). When using both indices, only regions from FA were selected. Figure 4 visualized the involved regions composed of the final combination.

To separate patients with PD from HC, the F1 score ranged from 70.4 to 80.7% with the highest score from the combination of FA or both indices (Table 2). In contrast, the sensitivity ranged from 66.7 to 80.7% for patients with PD, with the highest value from the combination of FA or both indices (Table 2).

Discussion

Main findings

This study proposed an artificial intelligence algorithm based on CNN. We demonstrated the potential in the diagnosis of PD by using diffusion tensor imaging. The results showed that the highest AUC is 0.941 by combining FA from 22 brain regions.

The MRI are often performed to rule out concomitant neurological disorders rather than confirm potential diagnoses. Diffusion tensor imaging can easily be incorporated into the routine imaging protocol. Image from the subject can be automatically parcellated. A CNN was applied to each parcellated brain region. An selective stacking algorithm, which combined the results from multiple regions, was implemented for the final diagnosis. The diagnostic performance can be improved when compared to the conventional ROC analysis in single region of the brain. The confidence from the clinician might be increased because image based examination such as diffusion tensor imaging can be less influenced by the fluctuations in the patient's condition.

The deep neural network contains many layers that transform the input to features with the convolution filter and nonlinearity. Usually only the final detection result is available. A lot of effort has been made for a transparent CNN (Eitel et al., 2019; Liu et al., 2020). In this study, images from parcellated brain regions were input to the neural network. The greedy algorithm was implemented which weighed the contribution from each ROI and determined the possibility of being a patient. Out result showed that the final diagnosis could be made by using FA from a limited number of ROIs, thus elucidated the involved regions and the possibility to interpret the finding clinically.

Convolutional neural network on Parkinson Disease

Artificial Intelligence has been previously used in the diagnosis of PD. Different approaches included practical usability constraints and deep learning (Hammerla et al., 2015), end-to-end 3D CNN (Esmaeilzadeh et al., 2018), or feature extraction and classification (Amoroso et al., 2018). Although the diagnostic accuracy varied between 93% (Amoroso et al., 2018) and 99% (Simonyan & Zisserman, 2014), these results were not validated using an independent dataset. Our algorithm combined the training of CNN and a selective stacking process. The blind validation demonstrated the validity of the approach in a clinical scenario.

The residual variation can be unknowingly extracted by a CNN as if that variation represented the underlying model structure. Deep neural networks that have been fine-tuned on the validation set are likely to overfit in real-world data if on a small-size dataset (Singh & Samavedham, 2015). It is common to utilize manually tuned data augmentation to increase the number of training examples (Akkus et al., 2017; Nowlan & Hinton, 2018; Oliveira et al., 2017; Pereira et al., 2016; Ronneberger et al., 2015). Additional approaches might stop training as soon as performance on a validation dataset starts to get worse (Caruana et al., 2001), or use weight penalties of various regularizations and soft weight sharing (Nair & Hinton, 2010). In this study, we employ the above methods in deep learning for PD detection for reducing redundant components to suppress the potential over-fitting. The result showed that the F1 in the cross validation varied from 84.4 ± 3.7% 89.7 ± 3.6%, compared to 80.7% in the blind validation. The diagnostic performance in blind validation are satisfactory with the cross validation. The difference might be attributed to the number of participants in our cohort, will be investigated in the future study.

Identified regions

Pathologically, PD is attributed to the death of dopaminergic neurons in the subcortical regions. However, in our study, the alterations were found to be in general brain regions of interest rather than in connecting regions, such as orbital part of middle frontal gyrus, occipital gyrus, calcarine, and cingulate gyrus. Significant atrophy has been reported

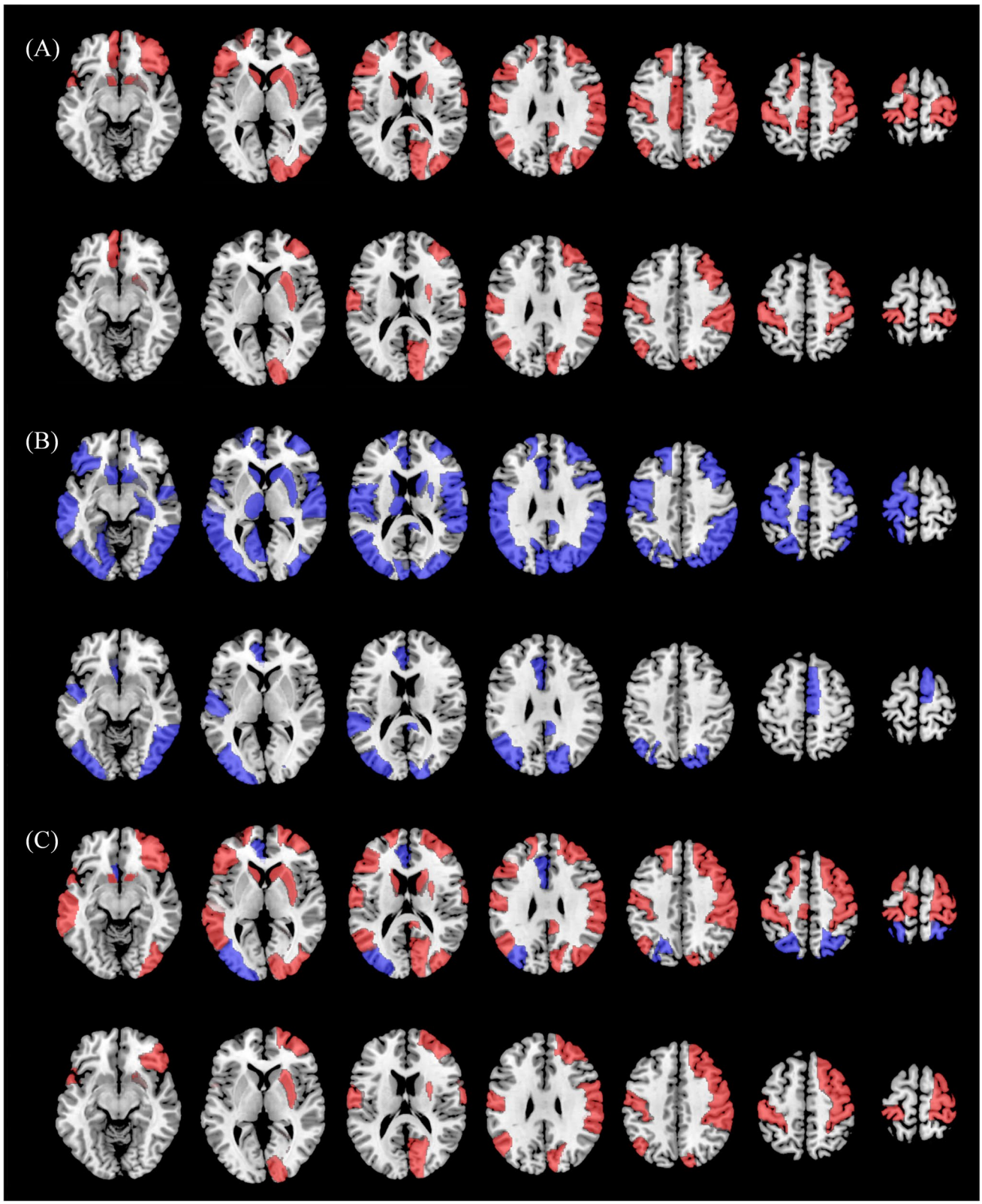

Fig. 4 The visualization of involved regions in the final combination with the highest AUC value in fractional anisotropy (FA), mean diffusivity (MD), and both indices. The figure is a visualization of the parcellated brain regions as in the combination by using (**A**) FA, Upper Panel: the final combination (22 regions). Lower Panel: fewer regions of combinations (9 regions), (**B**) MD, Upper Panel: the final combination (41 regions). Lower Panel: fewer regions of combinations (12 regions), and (**C**) both indices, Upper Panel: the final combination (27 regions). Lower Panel: few region of combination (12 regions)

Table 2 The diagnostic performance from the combination

	Index	HC	PD
TPR	FA	74.4%	80.7%
	MD	69.8%	66.7%
	FA+MD	74.4%	80.7%
TNR	FA	80.7%	74.4%
	MD	66.7%	69.8%
	FA+MD	80.7%	74.4%
PPV	FA	74.4%	80.7%
	MD	61.2%	74.5%
	FA+MD	74.4%	80.7%
NPV	FA	80.7%	74.4%
	MD	74.5%	61.2%
	FA+MD	80.7%	74.4%
ACC	FA	78.0%	78.0%
	MD	68.0%	68.0%
	FA+MD	78.0%	78.0%
F1	FA	74.4%	80.7%
	MD	65.2%	70.4%
	FA+MD	74.4%	80.7%

The table summarized the diagnostic performance from the blind dataset using the combination with reduced number of regions. HC, healthy control; PD, Parkinson's Disease; FA, fractional anisotropy; MD, mean diffusivity; TPR, true positive rate; TNR, true negative rate; PPV, positive predictive value; NPV, negative predictive value; ACC, accuracy; F1, harmonic mean of precision and sensitivity

in these regions in patients with PD (Ran et al., 2020; Xu et al., 2018). The decreased functional connectivity in orbital part of middle frontal gyrus was found in PD patients with impulse control disorders, which might be related to motor inhibition with prefrontal network (Gan et al., 2021).

Previous studies have demonstrated that loss of memory retrieval and storage is associated with alterations in the hippocampus (Calabresi et al., 2013; Hall et al., 2014), parahippocampus (Pagonabarraga et al., 2013), and amygdala (Vasconcellos et al., 2018) in patents with PD. Patients with PD who present with mild cognitive impairment or dementia exhibit significant atrophy in the angular gyrus, hippocampus, and putamen (Vasconcellos et al., 2018). In our study, the selection of the postcentral, putamen, and angular by the greedy algorithm may be associated with the in the non-motor symptoms of PD.

Although PD is characterized by specific motor disability, non-motor abnormalities may aggravate motor symptoms (Weil et al., 2016). The selection of the calcarine, cuneus, and angular may be associated with differences in visual malfunctions in PD. Indeed, previous fludeoxyglucose (FDG) uptake studies have revealed that patients with PD exhibit hypometabolism in cuneus, angular, orbital part of middle frontal gyrus, occipital gyrus and calcarine, which are associated with deterioration of cognitive function (Gonzalez-Redondo et al., 2014). Our findings may suggest that the diagnostic performance may be more strongly related to the extent of non-motor and cognitive impairment than the extent of motor abnormalities.

Limitation

The diffusion tensor index is susceptible to different acquisition parameters, including TE, TR, and resolution, scanner, and coil selection. As a result, fluctuation in the quantification of diffusion measurements might occur (Chen et al., 2019). Significant difference can be noticed from 129 to 180 features ($p<0.05$ with Bonferroni correction). Although such variations in acquisition parameters might increase the bias in the classification, our cross-validated study was further confirmed by an independent blind dataset. The result showed that even with 3 different imaging protocols, the diagnostic performance can be as good as 94.1±3.2%. However, this preliminary finding still required further validation using more data from a wide range of scanners in a multi-center study.

In either MD, FA, or the combined, we did not find any significant difference in the diagnostic performance between using the initial regions either with the highest AUC or random selection. With the result of AUC of 94.1±3.2%, we concluded that this process using regions with the highest AUC might lead to a reasonable diagnostic performance. To identify the global optimal, an exhaustive test over all possible solutions might be considered. However, it could take more than the C_n^{90} ($n\in\{1,2,\ldots,90\}$) calculation steps in each combination. To improve the diagnostic performance, it would be interesting to further explore these combinations in the future studies using a larger population.

Conclusions

The current study proposes the convolution neural network with diffusion tensor imaging of the whole brain, which could provide objective evidence for diagnosing healthy control and idiopathic PD with fair to good diagnostic performance.

Author contribution H.L.Z.: statistical analysis, writing the manuscript. C.C.T.: data processing, statistical analysis, and revision of the manuscript. M.Y.Z.: conception and design of study and writing the first draft. F. H.: data processing and statistical analysis. Y.L.C and Y.C.L: acquisition of clinical data. J.J.W. and Y.P.L: conception and design of study, study supervision, writing and revision of the manuscript, obtaining funding.

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Data availability Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval The study was approved by the Chang Gung medical foundation Institutional Review Board (Approved Number: 201601386B0D001) and conducted following the Declaration of Helsinki.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publication Not applicable.

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